



Garvan Institute
of Medical Research



UNSW
SYDNEY



Introduction to DNA METHYLATION and METHYLOME

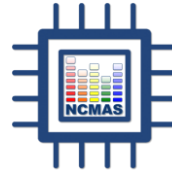
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All member of Prof. Susan Clark lab

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- Dr Braydon
- Dr Shalima



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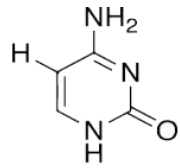


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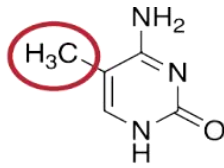
Outline

- Introduction to
- DNA methylation
- Profiling DNA methylation
- Whole Genome Bisulphite Sequencing (WGBS)
- Differential Methylated Region (DMR)
- IGV to visualize DNA methylation
- Enhancer and Promoter DNA methylation

What are DNA Methylation and genome-wide DNA methylation (Epigenome)?

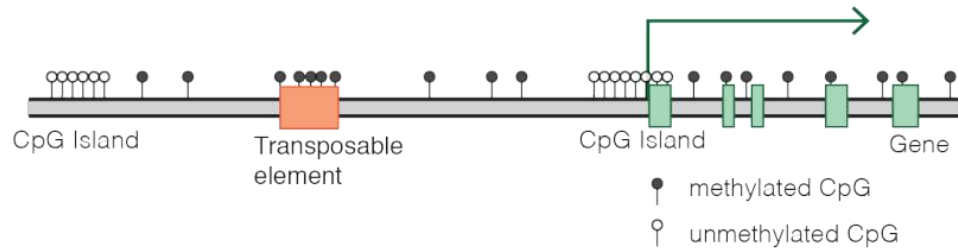
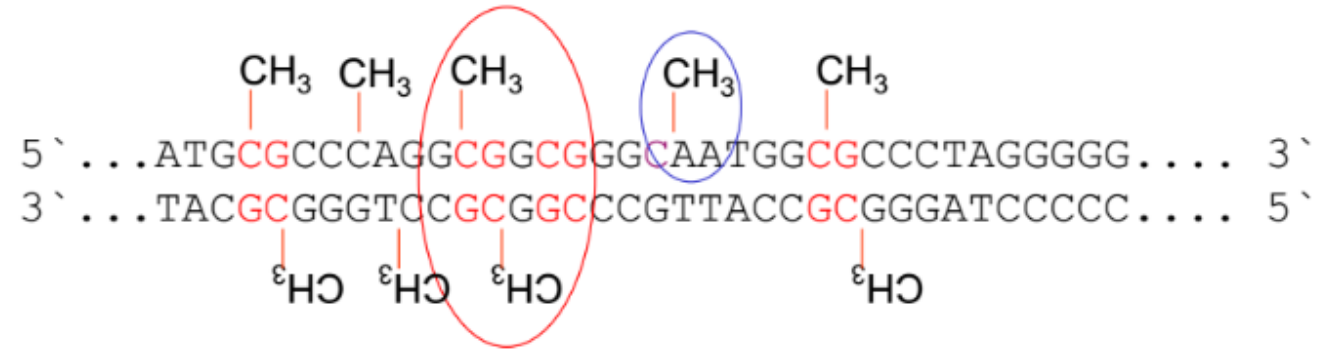


Cytosine

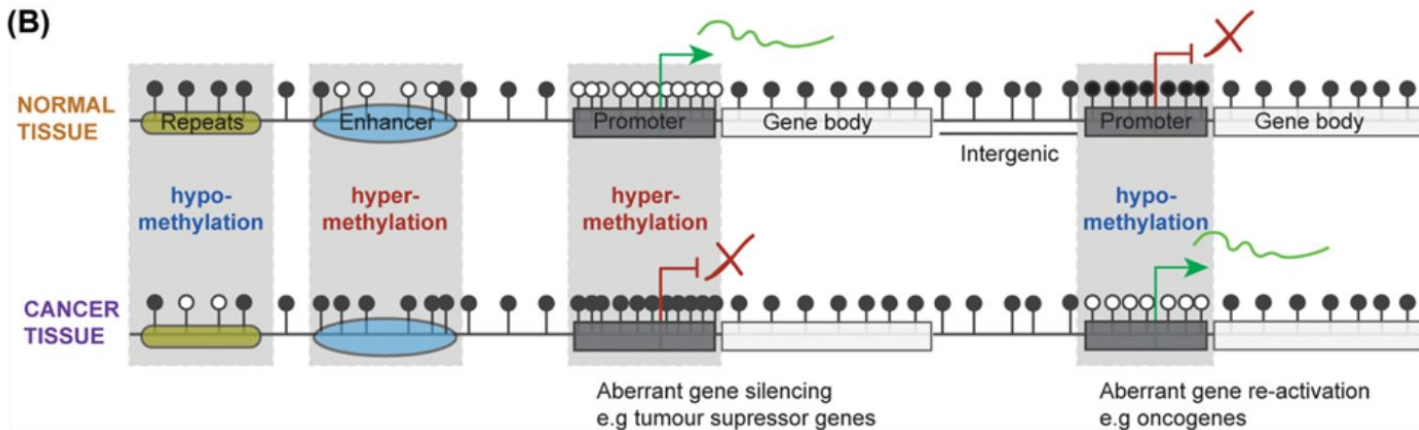


methyated Cytosine

Typical mammalian DNA methylation landscape

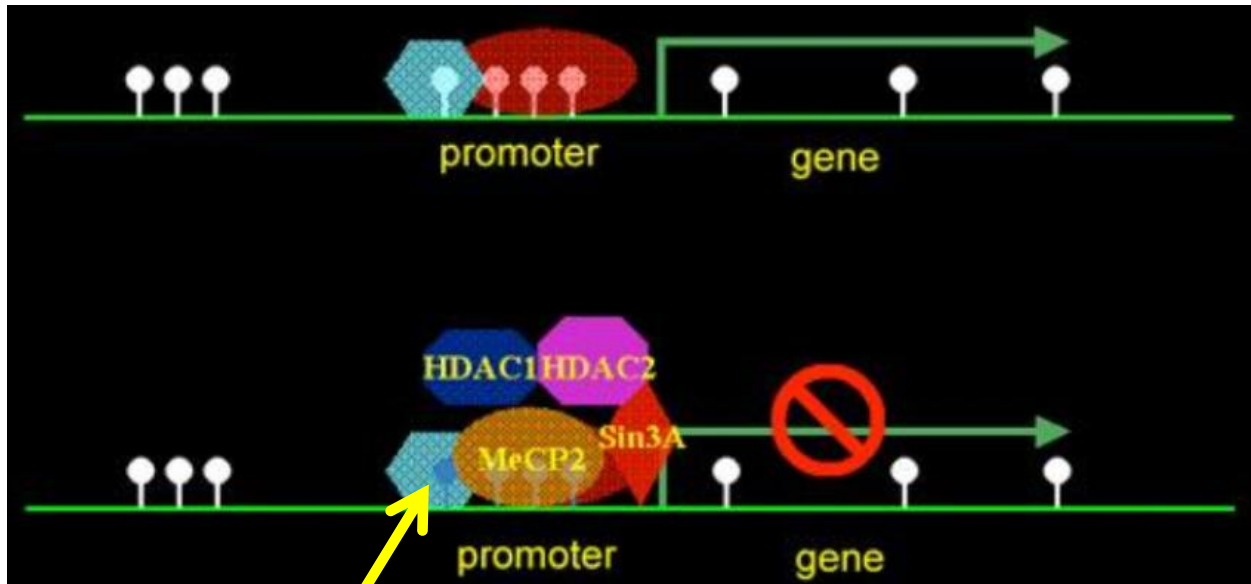


(B)



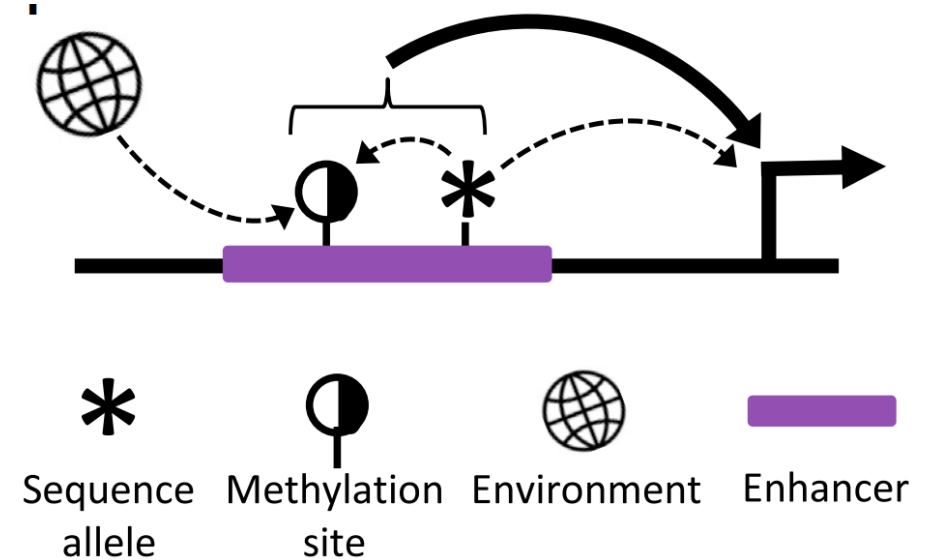
DNA methylation regulates gene expression

Promoter

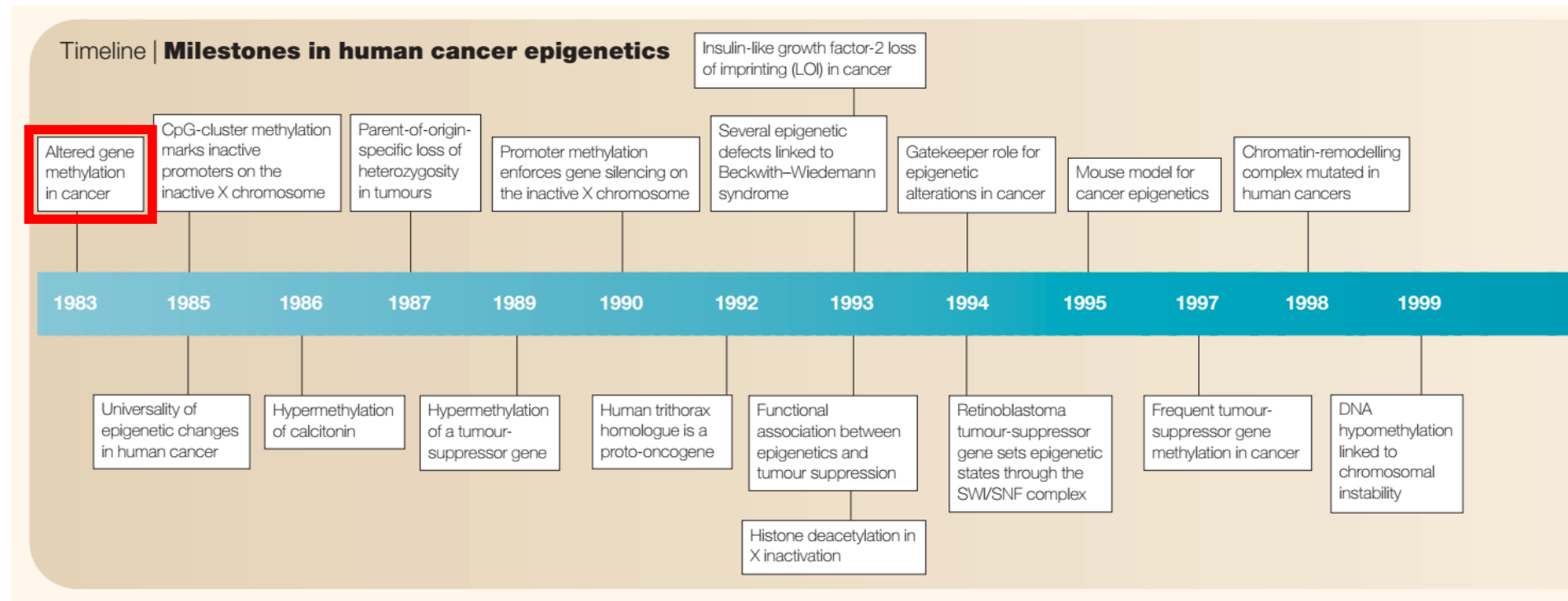


Methylated CpG site

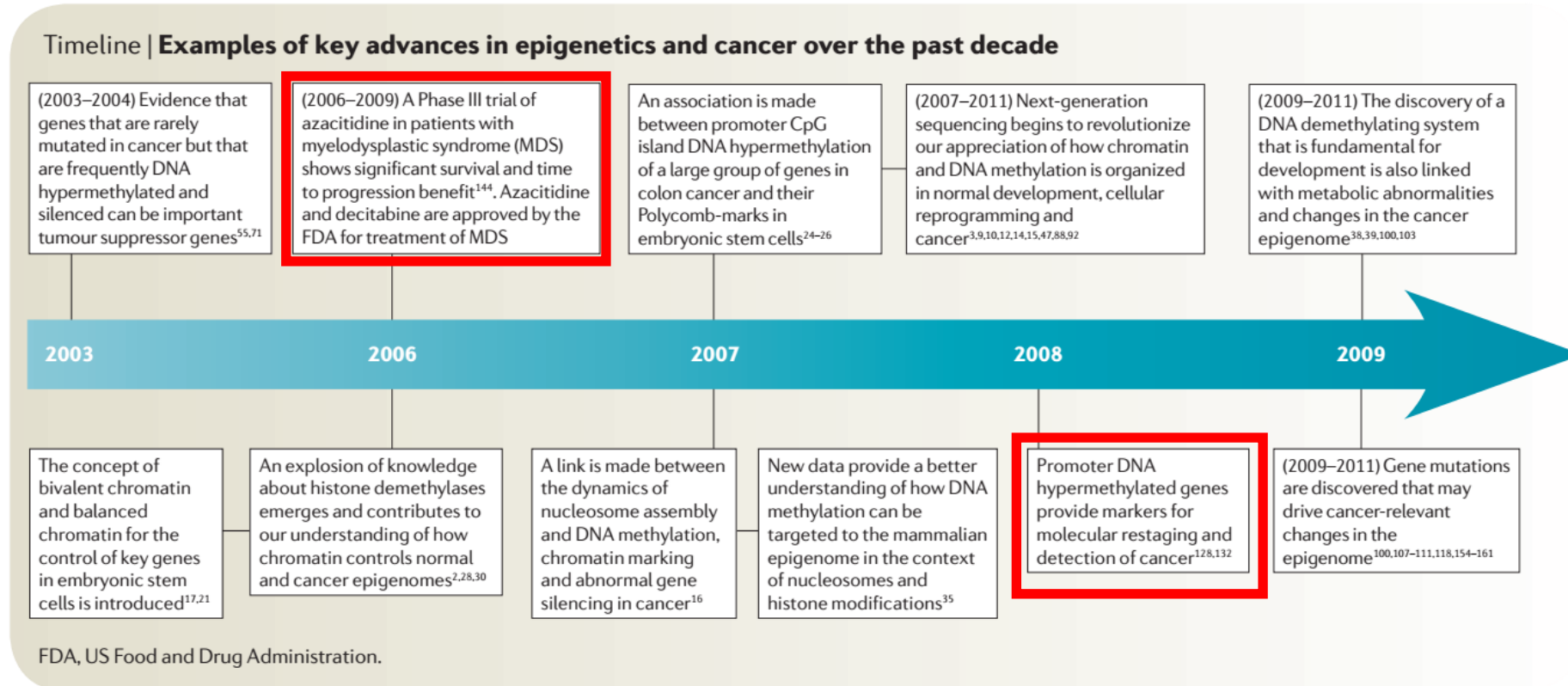
Enhancer



DNA methylation regulate gene expression in **cancer**

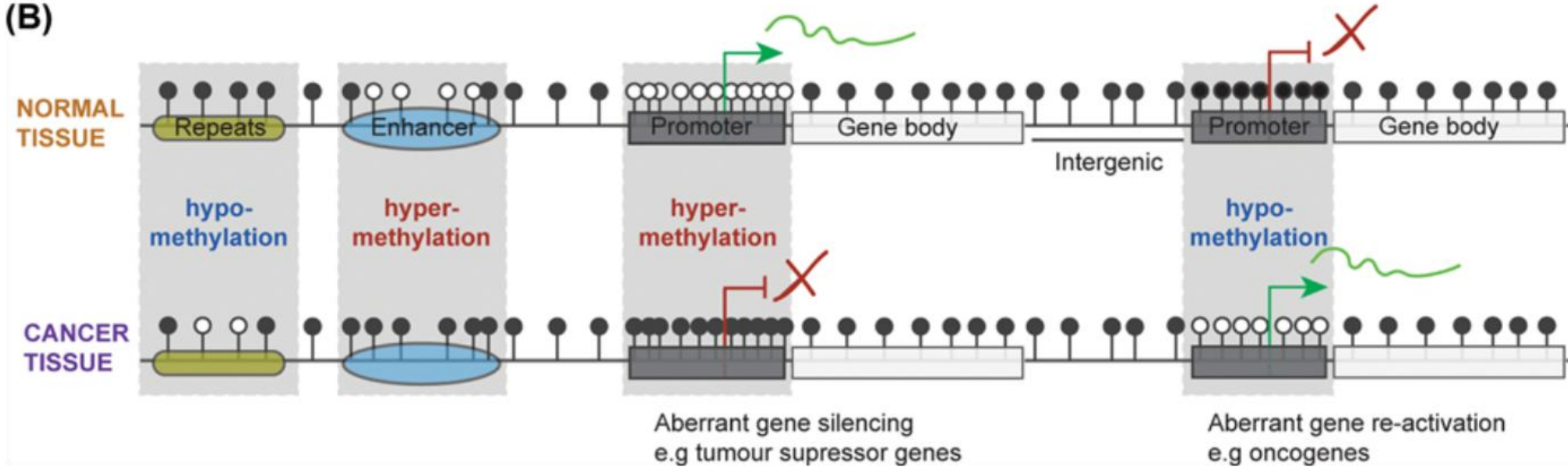


DNA methylation regulate gene expression in cancer

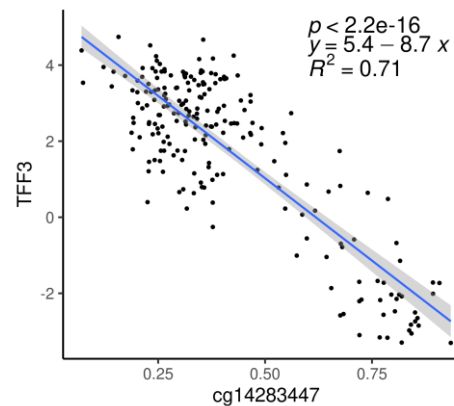
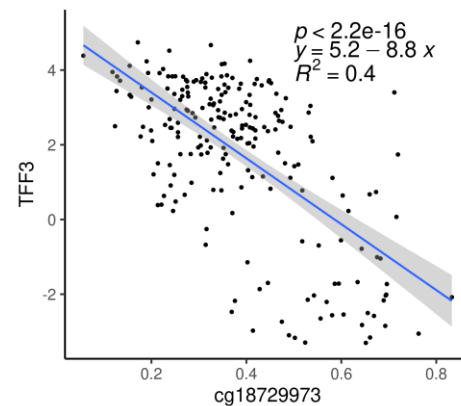


DNA methylation regulate gene expression in cancer

(B)

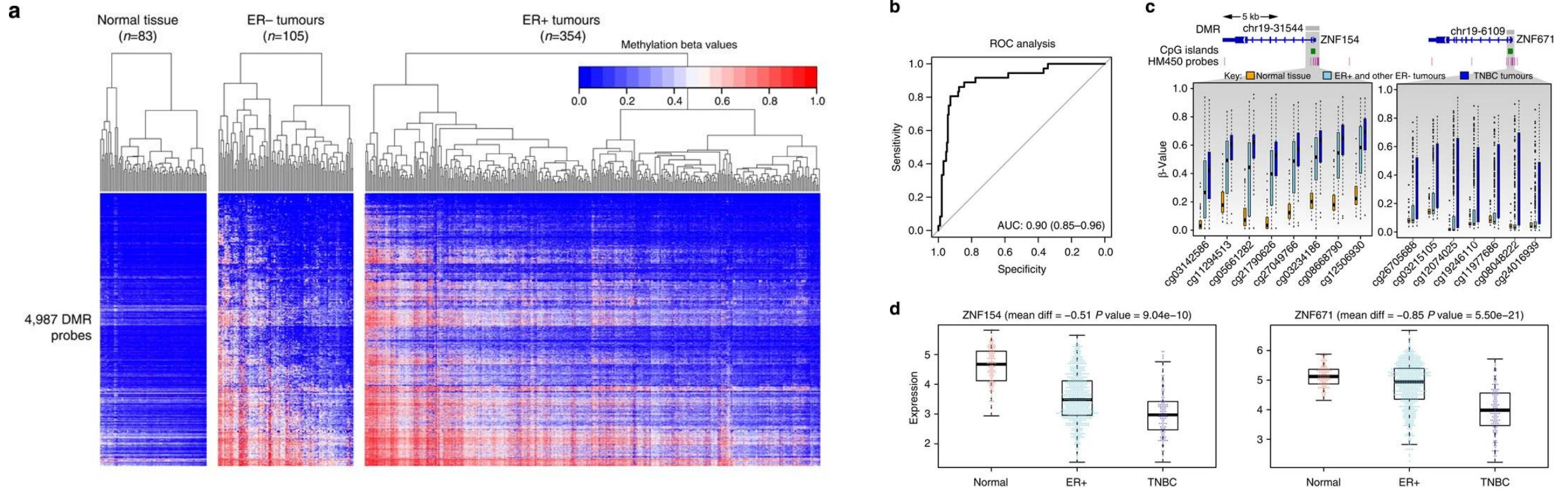


Expression



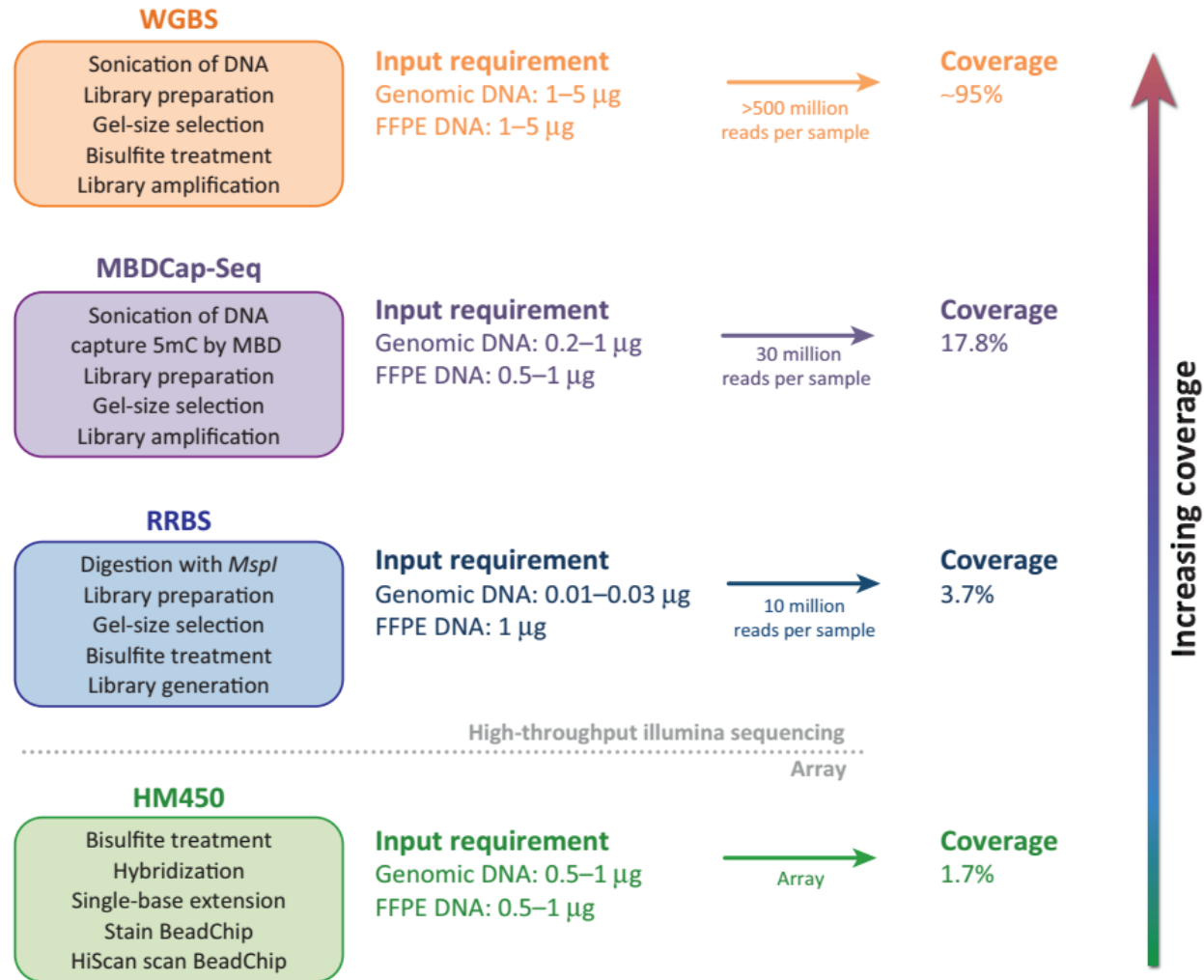
DNA methylation

DNA methylation regulate gene expression in breast cancer



Stirzaker C., Zotenko E., Song J.Z., Qu W., Nair S.S., Locke W.J., Stone A., Armstrong N.J., Robinson M.D., Dobrovic A... **SJ. Clark**. Methylome sequencing in triple-negative breast cancer reveals distinct methylation clusters with prognostic value. Nat. Commun. 2015

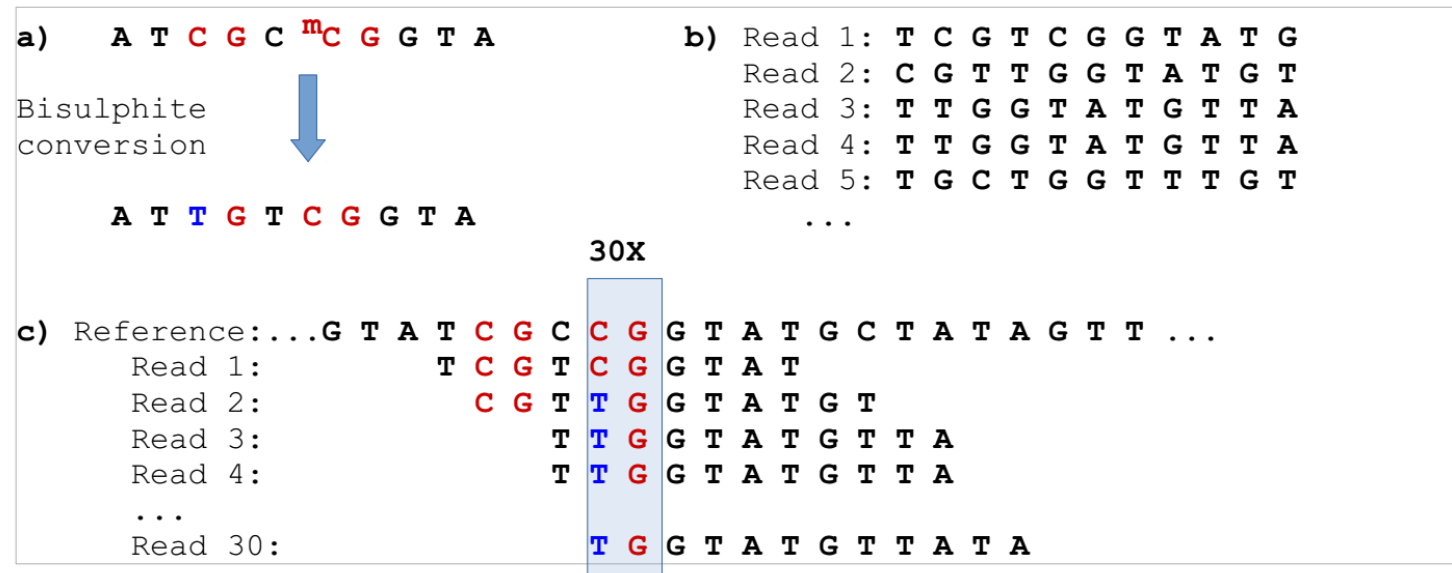
Methods to profile DNA methylation



850K/EPIC

10.1016/j.tig.2013.11.004

Bisulphite Sequencing



Beta value = Percent of Methylation (Perc)

$\text{Perc} = 100 \times \text{Meth} / \text{Cov}$

$\Rightarrow \text{Meth} = \text{Perc} \times \text{Cov} / 100$

$\Rightarrow \text{Cov} = \text{C} + \text{T}$

In the example:

$\text{Perc} = 100 \times 1 / 5 = 20\%$

$\text{Meth} = 1 = 20 \times 5 / 100$

$\text{Cov} = 1 + 4 = 5$

Figure 1. a) Sodium bisulphite converts unmethylated cytosines to thymines. b) The converted DNA is sequenced. c) Alignment and aggregation of these short sequence can be used to quantify the ratio of methylated (C) to unmethylated (converted T) cytosines from a population of cells at a single locus.

Total_cov_matrix_CpG.txt

	A	B	C	D	
1	chr	pos	nmDNA1	nmDNA2	
2	chr2	133031228	56	55	
3	chr2	133031240	74	81	
4	chr2	133031244	82	79	
5	chr2	133031249	64	70	
6	chr2	133031265	65	69	
7	chr2	133031268	67	56	

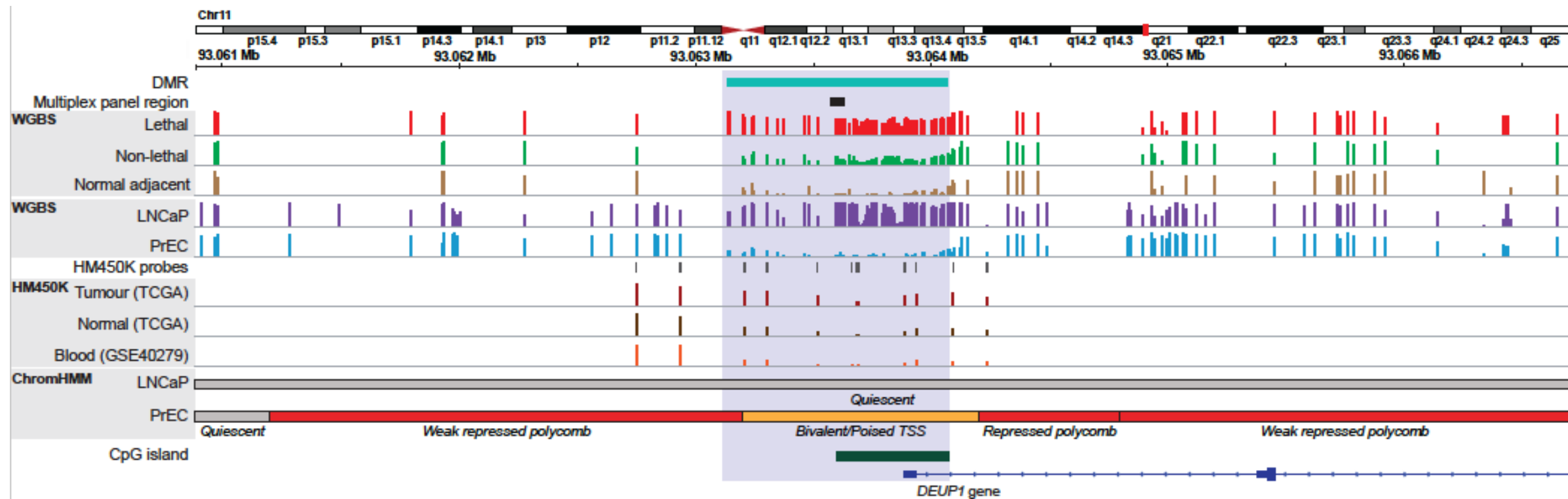
Meth_cov_matrix_CpG.txt

	A	B	C	D	
1	chr	pos	nmDNA1	nmDNA2	
2	chr2	133031228	41	28	
3	chr2	133031240	29	19	
4	chr2	133031244	47	47	
5	chr2	133031249	35	24	
6	chr2	133031265	19	21	
7	chr2	133031268	24	16	

Perc_cov_matrix_CpG.txt

	A	B	C	D	
1	chr	pos	nmDNA1	nmDNA2	
2	chr2	133031228	73	51	
3	chr2	133031240	39	23	
4	chr2	133031244	57	59	
5	chr2	133031249	55	34	
6	chr2	133031265	29	30	
7	chr2	133031268	36	29	

WGBS vs 450K



Quiz: Genetics vs Epigenetics

- Cancer: Hypermethylation of promoter regions (e.g., BRCA1, MLH1)
- PWS/Angelman: Imprinting errors (deletion or silencing of chromosome 15q11-q13)
- Type 2 Diabetes: Altered methylation in pancreatic cells and insulin-sensitive tissues
- Autoimmune (SLE): Global hypomethylation of DNA in immune T-cells

THANK YOU FOR YOUR ATTENTION!

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For further information!